Space group 166 as a maximal subgroup of space group 227

In this set of notes we show that the primitive lattice vectors of spacegroups 166 and 227 are compatible given a suitable choice, which we derive below.

Space group 166 has primitive lattice vectors given by:

$$\mathbf{R}_1^{p,166} = \frac{1}{2}(1, 1/\sqrt{3}, 2\alpha/3), \mathbf{R}_2^{p,166} = \frac{1}{2}(-1, 1/\sqrt{3}, 2\alpha/3), \mathbf{R}_3^{p,166} = \frac{1}{2}(0, -2/\sqrt{3}, 2\alpha/3)$$

Space group 227 has primitive lattice vectors given by:

$$\mathbf{R}_1^{p,227} = (0, 1/2, 1/2), \mathbf{R}_2^{p,227} = (1/2, 0, 1/2), \mathbf{R}_3^{p,227} = (1/2, 1/2, 0)$$

These lattice vectors have a pairwise dot product of $1/2$. As one may check, the lattice vectors for space group 166 will also have a pairwise dot product of $1/2$ if one chooses $\alpha = \sqrt{6}$, in which case we have:

$$\mathbf{R}_1^{p,166} = (1/2, 1/(2\sqrt{3}), \sqrt{2/3}), \mathbf{R}_2^{p,166} = (-1/2, 1/(2\sqrt{3}), \sqrt{2/3}),$$
$$\mathbf{R}_3^{p,166} = (0, -1/\sqrt{3}, \sqrt{2/3})$$

And clearly,

$$\mathbf{R}_1^{p,166} \cdot \mathbf{R}_2^{p,166} = -1/4 + 1/12 + 2/3 = (-3 + 1 + 8)/12 = 1/2,$$
$$\mathbf{R}_2^{p,166} \cdot \mathbf{R}_3^{166} = \mathbf{R}_3^{p,166} \cdot \mathbf{R}_1^{166} = -1/6 + 4/6 = 1/2$$